Day 6 Task — Trace a Request

Time needed	Deadline	Where to submit
45 minutes	Before the Day 7 session	Portal → Day 6 → Upload

Using only the tools already on your laptop, you will map the path from your machine to the internet. Open a terminal: Windows → search **cmd**; macOS → **Terminal**; Linux → Ctrl+Alt+T. Type each command, read the output, fill the table. Private IPs (192.168.x.x, 10.x.x.x) are fine to share; blur or omit your public IP if you prefer.

Step 1 — Who am I?

Windows: **ipconfig /all** · macOS: **ifconfig** (or System Settings → Wi-Fi → Details) · Linux: **ip a** and **ip r**

Field	Your value
Your private IPv4 address	
Subnet mask	
Default gateway (router) IP	
DNS server(s) your device uses	
MAC address of your Wi-Fi adapter (first 3 pairs only, e.g. 3C:5A:B4)	
Manufacturer of your adapter — look up those 3 pairs at macvendors.com	

Step 2 — Is the network alive? (ping)

Run each and record the average round-trip time in milliseconds (or 'no reply').

Command	Average time (ms)	What it tells you
ping <your gateway IP>		Can I reach my own router?
ping 8.8.8.8		Can I reach the internet by IP?
ping google.com		Does DNS + internet both work?

2D. If ping 8.8.8.8 works but ping google.com fails, what is broken? (1 line)

Step 3 — Which road does it take? (traceroute)

Windows: **tracert google.com** · macOS/Linux: **traceroute google.com**. Wait for it to finish (up to 30 hops; some show * * *).

Question	Your answer
How many hops to reach google.com?	
IP of hop 1 (should match your gateway)	
First hop that is NOT a private IP (your ISP's edge)	
Which hop had the biggest jump in time? Why might that be?	
How many hops showed * * * (no reply)? Why do routers do that?	

Step 4 — Ask the phone book (nslookup)

Run **nslookup** followed by the name. Record the IP address(es) returned and which DNS server answered.

Name	IP address(es) returned	DNS server that answered
google.com		
tekloomy.com		
your college or employer's website		

4D. Run **nslookup google.com 1.1.1.1** (ask Cloudflare's DNS instead). Did you get the same IPs? Why might two DNS servers answer differently? (2 lines)

Step 5 — Who is my laptop talking to? (netstat)

Windows: **netstat -ano** · macOS: **netstat -an | grep ESTABLISHED** · Linux: **ss -tunap**. Open your browser to a site first so you have live connections.

Question	Your answer
How many connections are ESTABLISHED?	
Pick one: remote IP and port (e.g. 142.250.x.x:443). What service is port 443?	
Any LISTENING ports on your machine? List up to 3 with their numbers.	
Look up one remote IP at ipinfo.io — which company owns it?	

Step 6 — Draw the path

On paper or in any drawing tool, draw the journey of one request from your laptop to google.com: laptop → Wi-Fi/router (gateway) → ISP → internet hops → Google. Label each box with the IP you found above, and mark where NAT happens and where DNS was consulted. Photograph or export it and include it in your submission.

Step 7 — One question (3–4 lines)

A colleague says ‘the website is down’. Using ping, nslookup and traceroute, describe the order you would check things in and what each result would tell you.

How to submit

- One file: Google Doc, PDF, or clear photos of this sheet filled in, plus your drawing
- Must contain: Steps 1–5 tables, the drawing, the Step 7 answer
- Upload under Day 6 on the portal before the Day 7 session

Marks (out of 10)

Part	Marks
Steps 1–5 completed with real values	5
Step 6 drawing, correctly labelled with NAT and DNS	2
Step 7 troubleshooting answer in the right order	3

Counts toward your lab journal (30%). Tomorrow we run the same commands inside your Ubuntu VM and see what they look like in Wireshark on Day 8.